



Progress Update 3: Segfault Slayer

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Game Overview

Game Idea:

- Metroidvania-style 2D side-scroller
- Player fights programming concepts and bugs (mainly C++ related)
- Playful, Pixel-theme (e.g. ASCII-Art for textures)

World Design:

- First Area: Hardware with Logic Operators
- Second Area: Programming with loops and other concepts

Previous Goals

- Player/Enemy Health System
- More Enemies / Boss
- Keyboard Binding Management
- Entity spawns in the Map
- Game Saves

Audio Manager

Centralized Audio Playback

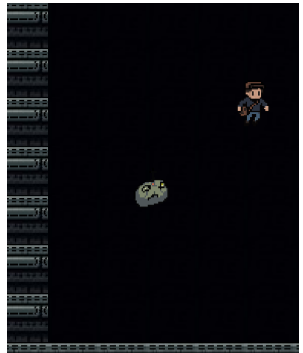
- Singleton AudioManager owns every sound and music
- constructed in similar fashion to AssetManager
- Buffers loaded once, lazily, on first use
- Master sound & music volume split - can be adjusted independently
- Global pause / resume / stop wired into pause menu

```
32 class AudioManager {
33     public:
34         [[nodiscard]] static AudioManager &getInstance();
35
36         ~AudioManager() = default;
37         AudioManager(const AudioManager &) = delete;
38         AudioManager &operator=(const AudioManager &) = delete;
39         AudioManager(AudioManager &&) = delete;
40         AudioManager &operator=(AudioManager &&) = delete;
41
42         void playSound(SoundEffect sfx, float volume = 100.f);
43         void pauseAllSounds();
44         void resumeAllSounds();
45         void stopAllSounds();
46
47         void playMusic(MusicTrack track, float volume = 30.f);
48         void stopMusic();
49         void pauseMusic();
50         void resumeMusic();
51         [[nodiscard]] MusicStatus musicStatus() const;
52 }
```

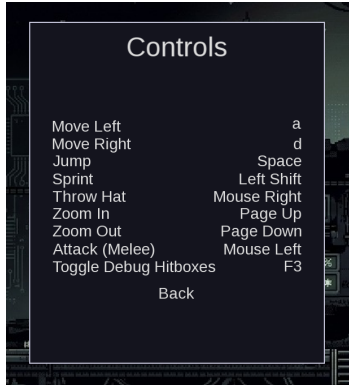
User Ability

Wall Jump

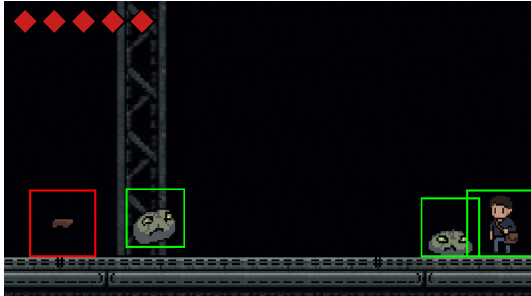
- Allows player to slide and jump from walls
- Key part in reaching second game area
- Obtainable from defeating first boss



Menus, Menus, Menus



Combat System



Overlap and Damage Resolver

- Entities publish Hitbox / Hurtbox rectangles each frame
- GameScene collects them and sends them to CombatSystem
- Central CombatSystem resolves every hitbox against every opposing-team hurtbox
- Each attack gets a `sourceId` - one (source, victim) pair damages at most once across all its active frames
- Invulnerable hurtboxes are skipped (i-frames during knockback)

Combat System Mechanics

- **Knockback**

Entities get short velocity boost in opposite direction of received hit.

- **Hurt-Flash**

Short red sprite coloring on taking damage.

- **i-Frames**

Short invincibility time when boxes first overlap. Allows for more fluid game feel.



Demo & Code Review

Next Steps

- More Enemies / Boss
- Inventory and State Menu
- Obtainable Player Abilities
- Map Expansion (Entity Spawns, Teleport Tiles, Collectables)
- Game Saves